ReLEAF Farmer's Guide

1. Welcome to ReLEAF

ReLEAF makes a liquid that helps protect your crops from stress like heat, drought and salty soil. The helpful bacteria stay sealed inside a cartridge. You never touch them. Only the protective liquid comes out. To make the whole process more sustainable, you can choose to purchase the medium that is made with agricultural waste, including hay, okara, and sugarcane bagasse, by collaborating with the farmer unions. (Refer to the sustainability page for more details)

Your job is simple:

1. Check for notifications in our app to see if there are any alerts.
2. Collect the protective liquid.
3. Notify ReLEAF members through the app when any help is needed.

Our job: we install the machine, deliver new cartridges and Nutrition medium packs, take the old ones back, and take the whole machine back if anything goes wrong. Anything that could carry live bacteria comes back to us. You never have to clean it or throw it away.

Words used in this guide

Word	What it means
Fill port	The opening on top where you pour in nutrition medium
Cartridge	The sealed part that holds the bacteria and the filter. Only we open it
Nutrition medium pack	A sealed bottle that keeps the medium balanced. We bring and swap it
Waste bottle	A sealed bottle where old medium goes after cleaning. We take it back
Collection bottle	Where the clear protective liquid collects. This is your product
Screen	The touch screen that shows readings, notices and buttons

Read Section 2 before you use ReLEAF for the first time.

2. Safety first

One rule covers almost everything: **if it is sealed, leave it sealed.** The cartridge, the nutrition medium pack and the waste bottle only ever come back to us closed.

Always

- Switch on and off in the right order (Section 4). The wrong order can break the control board.
- Keep water and spills away from the control board and wires on the top shelf.
- Keep the power plug where you can pull it out quickly.
- Use clean hands and a clean jug when you pour nutrition medium.

- Close the fill port cap after pouring.

Never

- Never open the cartridge, the nutrition medium pack or the waste bottle.
- Never unscrew tubes, sensors or connectors. Only the fill port, the collection bottle and the screen are yours to use.
- Never pour anything into the fill port except medium or bacteria cartridge.
- Never use ReLEAF liquid for food, drink, people or animals.
- Never change or unplug any wire while the power is on.
- Never put your fingers near the moving valves or the pump while it is running.

If something goes wrong

What happens	What to do
Liquid leaking from the machine	The bioreactor already has multiple existing failsafe mechanisms, bbut in the case of leakage, the bioreactor will automatically stop running.
Burning smell, smoke or a hot board	Switch off the power first, then unplug the board. Do not touch until cool. Call us
Collection liquid looks cloudy	Press **Emergency Stop**. Do not use the liquid. Call us, and we will take the machine back.
Liquid from inside the machine on skin	Wash with soap and water
Liquid from inside the machine in the eyes	Rinse with clean water for 15 minutes. See a doctor if it still hurts
A red warning bar on the screen	Follow Section 8

When in doubt, stop the machine and call us. We will collect it — you never need to clean or repair it yourself.

3. The parts you'll use

We install everything. These are the only parts you need to know.

Part	What it does	You will...
Touch screen	Shows readings, notices and buttons	Look at it every day
Fill port	Opening for agricultural waste medium	Pour in fresh medium when asked
Collection bottle	Collects your protective liquid	Empty it before it fills
Power switch and plug	Powers the machine	Switch on and off in order

Part	What it does	You will...
Cartridge	Holds the bacteria and filter	Leave it alone. We swap it
Nutrition medium pack	Keeps the medium balanced	Leave it alone. We swap it

Everything else (pumps, valves, tubes, sensors, wires) is ours to look after.

4. Setting up

We install and test the machine for you, put the first cartridge and nutrition medium pack in, and show you how to use the screen.

Keep it in a good spot

- Indoors, out of direct sun and rain, on a flat, steady surface.
- Keep the area around and under it clear, so you can see any leaks.
- Keep the power plug easy to reach.
- Do not move the machine yourself. Call us if it needs to move.

Switching on (do in this order)

1. Plug in the control board's USB-C cable first.
2. Wait until the screen has fully started.
3. Then switch on the 12 V power supply.

Switching off (reverse order)

1. Switch off the 12 V power supply first.
2. Then unplug the control board.

If the screen does not start: unplug the board, wait 10 seconds, plug it back in, and follow the ON order again. If it still does not start, call us.

Checking the screen

The status line at the top must show **Connected** and **MCU: OK**. If it does not, do not use the machine — call us. The screen can be switched between English and Traditional Chinese with the language button at the top right.

5.Purchase nutrition medium

To prepare the medium for culturing the bacteria, please directly purchase the TB medium or nutrition medium made by agriculture waste from ReLEAF.

6. Daily checks

Once a day, spend two minutes on these checks. The machine looks after everything else by itself.

Check	Good	If not...
Floor and surface under the machine	Dry	Press **Emergency Stop** and call us
Screen status line	**Connected** and **MCU: OK**	Call us
Notices on the screen	None	Follow the notice (Section 8)
Flow reading	Moving, about the same as yesterday	Section 11
Collection bottle	Less than three-quarters full	Empty it (Section 7)
Collection liquid	Clear	Cloudy: press **Emergency Stop** and call us

You do not need to top up, adjust pH or touch any settings. When the medium is used up, the screen will tell you (Section 9).

7. Collecting the protectant

The clear liquid in the collection bottle is your product. Every cartridge is tested by us before delivery, so the filter is checked before it reaches you.

Emptying the collection bottle

1. On the screen, press **Close collection**. Wait for the valve to close.
2. A ReLEAF member will assist with protectant bottle collection
3. Press **Open collection**.

Good liquid vs. bad liquid

- Good: clear, with nothing floating in it.
- Bad: cloudy, or the screen shows a **Filter alarm**. Press **Emergency Stop**, do not use the liquid, keep the bottle capped and call us. We will collect the bottle and the machine.

Using it on your crops

Apply ReLEAF protectant only as shown on the product label and only where it is approved for use in your area. The protectant should be applied 9 times every 120 days, and 5100 Liters of ACCD culture should be applied per hectare per application, at a dilution ratio of 1 part stock, and 1 part diluent.

8. Notices and warnings on the screen

The machine protects itself. When something is wrong it stops the pumps and shows a notice. Some notices are yours to act on; the rest mean "call us".

Notice	What it means	What you do
Time to change medium	The agricultural waste medium is used up	Notify ReLEAF members for assistance
Collection bottle full	The bottle needs emptying	Notify ReLEAF members for assistance
Cartridge due soon	Your next cartridge swap is coming up	Nothing. We will contact you to book a delivery
Nutrition medium pack low	The balancing nutrition medium is running out	Nothing. We will bring a new pack
Waste bottle full	Old medium needs collecting	Notify ReLEAF members to collect the waste bottle
Red **Pressure** warning	Something is blocked	The bioreactor will automatically stop running
Filter alarm	The filter may be damaged	The bioreactor will automatically stop running
Cleaning	The bioreactor needs to be cleaned	Notify ReLEAF members to assist you in cleaning the bioreactor
Any other warning	—	Call us. Do not try to fix it yourself

Never change settings or limits to make a warning go away.

9. Changing the medium and cleaning the bioreactor

When the screen shows **Time to change medium**, the machine needs to have the old medium cleaned out. You never drain, pour out or touch the old medium.

Step by step

1. Empty the collection bottle (Section 7).
2. Notify ReLEAF members through the app to help clean the bioreactor

Storing the machine when not in use

- Wipe only the outside with a dry or damp cloth. Keep water off the screen and board.
- Keep it dry and out of the sun.
- If you will not use it for more than a year, call us — we may collect the cartridge.

10. Cartridge swaps and take-back

We deliver new parts and take the old ones back on the same visit. It is always a one-for-one swap.

When we deliver...	We take back...
A new cartridge	The used cartridge, still sealed

When we deliver...	We take back...
A new nutrition medium pack	The used nutrition mediumfw pack, still sealed
An empty waste bottle	The full waste bottle, still sealed
A replacement machine	The broken machine

Before a delivery visit

- Do not take anything out yourself. Leave the used cartridge, pack and bottle in the machine. We remove them.

When the machine stops working

Press **Emergency Stop**, switch off in order, and call us. We will collect the whole machine, clean it, and bring a working one. Never throw away any part of ReLEAF yourself — not even the screen or the wires.

11. Problem solver

If your problem is not here, or the fix does not work: press **Emergency Stop**, switch off in order, and call us.

Problem	Try this
Screen does not start	Unplug the board for 10 seconds. Plug in the board first, then the power. Still dead: call us
Buttons do nothing	Switch off in order, wait 10 seconds, switch on again. Still stuck: call us
Flow reading shows zero	Check the screen for a notice. If medium was just added, wait 5 minutes. Still zero: call us
Ready for new medium never appears after cleaning	Do not open the fill port. Call us
Medium spilled while pouring	Medium is just food — wipe it up normally. Spill from inside the machine: call us
Collection bottle fills slowly	Normal as the cartridge ages. If it stops completely, call us
Collection liquid is cloudy	**Emergency Stop**, do not use it, call us
Any leak from the machine	**Emergency Stop**, switch off, call us

12. Frequently asked questions

How do I replace the medium? Wait for the **Time to change medium** notice. Contact ReLEAF members to order the new medium. Then pour the new medium into the fill port and press **Start**.

Where does the old medium go? ReLEAF members will keep the bioreactor safe by pumping it into the sealed waste bottle. We will collect the waste bottle on our next visit. Never open it.

How do I replace the cartridge? You don't. When the screen shows **Cartridge due soon**, we contact you and bring a new one. We take the old cartridge out and put the new one in on the same visit.

How do I replace the nutrition medium pack? You don't. When it runs low, we bring a new one and take the old one back.

Do I ever need to handle bacteria or chemicals? No. Everything that could carry live bacteria, and every chemical, stays sealed and comes back to us.

What if the machine breaks? The bioreactor will automatically stop. Switch off the machine and call us. We take the whole machine back to clean and repair it.

What do I do with the protective liquid? Use it on your crops as shown on the product label (Section 7).

13. Getting help and record sheet

Machine record (installer fills this in)

Item	Value
Date installed	
Machine number	
Screen address	
First cartridge number	
Installed by	

Swap log (we fill this in at each visit)

Date	New cartridge number	Old cartridge taken back?	Nutrition medium pack swapped?	Waste bottle taken back?

This guide is a simplified version of the ReLEAF Bioreactor User Manual. Technicians should use the full manual.